

GTOS Detailed Volume - Volume 08

Engagement Communication

Architecture

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- Notification — Orientation, Ownership, Consent, and Channel Capability
- Notification — Messages, Conversations, Templates, and State
- Notification — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Notification — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- WhatsApp — Orientation, Ownership, Consent, and Channel Capability
- WhatsApp — Messages, Conversations, Templates, and State
- WhatsApp — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- WhatsApp — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- WhatsApp Connector — Orientation, Ownership, Consent, and Channel Capability
- WhatsApp Connector — Messages, Conversations, Templates, and State
- WhatsApp Connector — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- WhatsApp Connector — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Instagram — Orientation, Ownership, Consent, and Channel Capability
- Instagram — Messages, Conversations, Templates, and State
- Instagram — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Instagram — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Email — Orientation, Ownership, Consent, and Channel Capability
- Email — Messages, Conversations, Templates, and State
- Email — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Email — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- WebSocket and Realtime Messaging — Orientation, Ownership, Consent, and Channel Capability
- WebSocket and Realtime Messaging — Messages, Conversations, Templates, and State
- WebSocket and Realtime Messaging — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- WebSocket and Realtime Messaging — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

- Conversation Ownership — Orientation, Ownership, Consent, and Channel Capability
- Conversation Ownership — Messages, Conversations, Templates, and State
- Conversation Ownership — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Conversation Ownership — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Consent and Channel Capability — Orientation, Ownership, Consent, and Channel Capability
- Consent and Channel Capability — Messages, Conversations, Templates, and State
- Consent and Channel Capability — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Consent and Channel Capability — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Template and Localization — Orientation, Ownership, Consent, and Channel Capability
- Template and Localization — Messages, Conversations, Templates, and State
- Template and Localization — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Template and Localization — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Human Handoff and AI-Assisted Response — Orientation, Ownership, Consent, and Channel Capability
- Human Handoff and AI-Assisted Response — Messages, Conversations, Templates, and State
- Human Handoff and AI-Assisted Response — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation
- Human Handoff and AI-Assisted Response — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests
- Engagement Channel Ownership and Consent Matrix

Notification — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-01-00
Subject	NotificationChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Notification
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Notification**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Notification**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
NotificationChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
NotificationChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateNotificationChannel
- ValidateNotificationChannel
- ApproveNotificationChannel
- ActivateNotificationChannel
- SuspendNotificationChannel
- CorrectNotificationChannel
- RetireNotificationChannel

Reference Events:

- NotificationChannelCreated
- NotificationChannelValidated
- NotificationChannelApproved
- NotificationChannelActivated
- NotificationChannelSuspended
- NotificationChannelCorrected
- NotificationChannelRetired

Reference queries:

- GetNotificationChannel
- ListNotificationChannelRecords
- GetNotificationChannelTimeline
- GetNotificationChannelEvidence
- GetNotificationChannelDecisionPackage
- GetNotificationChannelExceptions
- GetNotificationChannelAuditTrail

Reference API surface:

```
POST /api/v1/notificationchannel/commands
GET  /api/v1/notificationchannel/{id}
GET  /api/v1/notificationchannel/{id}/timeline
GET  /api/v1/notificationchannel/{id}/evidence
GET  /api/v1/notificationchannel/{id}/decisions
POST /api/v1/notificationchannel/{id}/exceptions
POST /api/v1/notificationchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
notificationchannel_aggregate	authoritative subject
notificationchannel_version	immutable submitted, approved, issued, or signed versions
notificationchannel_relationship	governed relationships
notificationchannel_decision	binding Decisions and rationale
notificationchannel_evidence_link	provenance and evidence relationships
notificationchannel_event_outbox	transactional Event publication
notificationchannel_idempotency	duplicate-write protection
notificationchannel_exception	exception, claim, dispute, or hold
notificationchannel_correction	correction and supersession lineage
notificationchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Notification — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-01-01
Subject	NotificationMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Notification
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Notification**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
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The accountable owner is **Engagement Architecture / Notification**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

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- create and identify the subject
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1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
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2. Ubiquitous Language and Subject Model

Subject	Required interpretation
NotificationMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
NotificationMessageModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
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- typed contracts
- immutable binding versions

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- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

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A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
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- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
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```
stateDiagram-v2
    [*] --> PROPOSED
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    VALIDATION_PENDING --> APPROVAL_PENDING
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```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
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5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

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Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateNotificationMessageModel
- ValidateNotificationMessageModel
- ApproveNotificationMessageModel
- ActivateNotificationMessageModel
- SuspendNotificationMessageModel
- CorrectNotificationMessageModel
- RetireNotificationMessageModel

Reference Events:

- NotificationMessageModelCreated
- NotificationMessageModelValidated
- NotificationMessageModelApproved
- NotificationMessageModelActivated
- NotificationMessageModelSuspended
- NotificationMessageModelCorrected
- NotificationMessageModelRetired

Reference queries:

- GetNotificationMessageModel
- ListNotificationMessageModelRecords
- GetNotificationMessageModelTimeline
- GetNotificationMessageModelEvidence
- GetNotificationMessageModelDecisionPackage
- GetNotificationMessageModelExceptions
- GetNotificationMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/notificationmessagemodel/commands
GET  /api/v1/notificationmessagemodel/{id}
GET  /api/v1/notificationmessagemodel/{id}/timeline
GET  /api/v1/notificationmessagemodel/{id}/evidence
GET  /api/v1/notificationmessagemodel/{id}/decisions
POST /api/v1/notificationmessagemodel/{id}/exceptions
POST /api/v1/notificationmessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
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Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

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Reference table families:

Table family	Purpose
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notificationmessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

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- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
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- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Notification — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-01-02
Subject	NotificationProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Notification
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Notification.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Notification**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
NotificationProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
NotificationProviderAdapter {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateNotificationProviderAdapter
- ValidateNotificationProviderAdapter
- ApproveNotificationProviderAdapter
- ActivateNotificationProviderAdapter
- SuspendNotificationProviderAdapter
- CorrectNotificationProviderAdapter
- RetireNotificationProviderAdapter

Reference Events:

- NotificationProviderAdapterCreated
- NotificationProviderAdapterValidated
- NotificationProviderAdapterApproved
- NotificationProviderAdapterActivated
- NotificationProviderAdapterSuspended
- NotificationProviderAdapterCorrected
- NotificationProviderAdapterRetired

Reference queries:

- GetNotificationProviderAdapter
- ListNotificationProviderAdapterRecords
- GetNotificationProviderAdapterTimeline
- GetNotificationProviderAdapterEvidence
- GetNotificationProviderAdapterDecisionPackage
- GetNotificationProviderAdapterExceptions
- GetNotificationProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/notificationprovideradapter/commands
GET  /api/v1/notificationprovideradapter/{id}
GET  /api/v1/notificationprovideradapter/{id}/timeline
GET  /api/v1/notificationprovideradapter/{id}/evidence
GET  /api/v1/notificationprovideradapter/{id}/decisions
POST /api/v1/notificationprovideradapter/{id}/exceptions
POST /api/v1/notificationprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
notificationprovideradapter_aggregate	authoritative subject
notificationprovideradapter_version	immutable submitted, approved, issued, or signed versions
notificationprovideradapter_relationship	governed relationships
notificationprovideradapter_decision	binding Decisions and rationale
notificationprovideradapter_evidence_link	provenance and evidence relationships
notificationprovideradapter_event_outbox	transactional Event publication
notificationprovideradapter_idempotency	duplicate-write protection
notificationprovideradapter_exception	exception, claim, dispute, or hold
notificationprovideradapter_correction	correction and supersession lineage
notificationprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Notification — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-01-03
Subject	NotificationOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Notification
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Notification.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Notification**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
NotificationOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
NotificationOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
NotificationOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateNotificationOperatingProfile
- ValidateNotificationOperatingProfile
- ApproveNotificationOperatingProfile
- ActivateNotificationOperatingProfile
- SuspendNotificationOperatingProfile
- CorrectNotificationOperatingProfile
- RetireNotificationOperatingProfile

Reference Events:

- NotificationOperatingProfileCreated
- NotificationOperatingProfileValidated
- NotificationOperatingProfileApproved
- NotificationOperatingProfileActivated
- NotificationOperatingProfileSuspended
- NotificationOperatingProfileCorrected
- NotificationOperatingProfileRetired

Reference queries:

- GetNotificationOperatingProfile
- ListNotificationOperatingProfileRecords
- GetNotificationOperatingProfileTimeline
- GetNotificationOperatingProfileEvidence
- GetNotificationOperatingProfileDecisionPackage
- GetNotificationOperatingProfileExceptions
- GetNotificationOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/notificationoperatingprofile/commands
GET  /api/v1/notificationoperatingprofile/{id}
GET  /api/v1/notificationoperatingprofile/{id}/timeline
GET  /api/v1/notificationoperatingprofile/{id}/evidence
GET  /api/v1/notificationoperatingprofile/{id}/decisions
POST /api/v1/notificationoperatingprofile/{id}/exceptions
POST /api/v1/notificationoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
notificationoperatingprofile_aggregate	authoritative subject
notificationoperatingprofile_version	immutable submitted, approved, issued, or signed versions
notificationoperatingprofile_relationship	governed relationships
notificationoperatingprofile_decision	binding Decisions and rationale
notificationoperatingprofile_evidence_link	provenance and evidence relationships
notificationoperatingprofile_event_outbox	transactional Event publication
notificationoperatingprofile_idempotency	duplicate-write protection
notificationoperatingprofile_exception	exception, claim, dispute, or hold
notificationoperatingprofile_correction	correction and supersession lineage
notificationoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-02-00
Subject	WhatsAppChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for WhatsApp**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppChannel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppChannel
- ValidateWhatsAppChannel
- ApproveWhatsAppChannel
- ActivateWhatsAppChannel
- SuspendWhatsAppChannel
- CorrectWhatsAppChannel
- RetireWhatsAppChannel

Reference Events:

- WhatsAppChannelCreated
- WhatsAppChannelValidated
- WhatsAppChannelApproved
- WhatsAppChannelActivated
- WhatsAppChannelSuspended
- WhatsAppChannelCorrected
- WhatsAppChannelRetired

Reference queries:

- GetWhatsAppChannel
- ListWhatsAppChannelRecords
- GetWhatsAppChannelTimeline
- GetWhatsAppChannelEvidence
- GetWhatsAppChannelDecisionPackage
- GetWhatsAppChannelExceptions
- GetWhatsAppChannelAuditTrail

Reference API surface:

```
POST /api/v1/whatsappchannel/commands
GET  /api/v1/whatsappchannel/{id}
GET  /api/v1/whatsappchannel/{id}/timeline
GET  /api/v1/whatsappchannel/{id}/evidence
GET  /api/v1/whatsappchannel/{id}/decisions
POST /api/v1/whatsappchannel/{id}/exceptions
POST /api/v1/whatsappchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappchannel_aggregate	authoritative subject
whatsappchannel_version	immutable submitted, approved, issued, or signed versions
whatsappchannel_relationship	governed relationships
whatsappchannel_decision	binding Decisions and rationale
whatsappchannel_evidence_link	provenance and evidence relationships
whatsappchannel_event_outbox	transactional Event publication
whatsappchannel_idempotency	duplicate-write protection
whatsappchannel_exception	exception, claim, dispute, or hold
whatsappchannel_correction	correction and supersession lineage
whatsappchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-02-01
Subject	WhatsAppMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for WhatsApp**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppMessageModel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppMessageModel
- ValidateWhatsAppMessageModel
- ApproveWhatsAppMessageModel
- ActivateWhatsAppMessageModel
- SuspendWhatsAppMessageModel
- CorrectWhatsAppMessageModel
- RetireWhatsAppMessageModel

Reference Events:

- WhatsAppMessageModelCreated
- WhatsAppMessageModelValidated
- WhatsAppMessageModelApproved
- WhatsAppMessageModelActivated
- WhatsAppMessageModelSuspended
- WhatsAppMessageModelCorrected
- WhatsAppMessageModelRetired

Reference queries:

- GetWhatsAppMessageModel
- ListWhatsAppMessageModelRecords
- GetWhatsAppMessageModelTimeline
- GetWhatsAppMessageModelEvidence
- GetWhatsAppMessageModelDecisionPackage
- GetWhatsAppMessageModelExceptions
- GetWhatsAppMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/whatsappmessagemodel/commands
GET  /api/v1/whatsappmessagemodel/{id}
GET  /api/v1/whatsappmessagemodel/{id}/timeline
GET  /api/v1/whatsappmessagemodel/{id}/evidence
GET  /api/v1/whatsappmessagemodel/{id}/decisions
POST /api/v1/whatsappmessagemodel/{id}/exceptions
POST /api/v1/whatsappmessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappmessagemodel_aggregate	authoritative subject
whatsappmessagemodel_version	immutable submitted, approved, issued, or signed versions
whatsappmessagemodel_relationship	governed relationships
whatsappmessagemodel_decision	binding Decisions and rationale
whatsappmessagemodel_evidence_link	provenance and evidence relationships
whatsappmessagemodel_event_outbox	transactional Event publication
whatsappmessagemodel_idempotency	duplicate-write protection
whatsappmessagemodel_exception	exception, claim, dispute, or hold
whatsappmessagemodel_correction	correction and supersession lineage
whatsappmessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-02-02
Subject	WhatsAppProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for WhatsApp**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppProviderAdapter
- ValidateWhatsAppProviderAdapter
- ApproveWhatsAppProviderAdapter
- ActivateWhatsAppProviderAdapter
- SuspendWhatsAppProviderAdapter
- CorrectWhatsAppProviderAdapter
- RetireWhatsAppProviderAdapter

Reference Events:

- WhatsAppProviderAdapterCreated
- WhatsAppProviderAdapterValidated
- WhatsAppProviderAdapterApproved
- WhatsAppProviderAdapterActivated
- WhatsAppProviderAdapterSuspended
- WhatsAppProviderAdapterCorrected
- WhatsAppProviderAdapterRetired

Reference queries:

- GetWhatsAppProviderAdapter
- ListWhatsAppProviderAdapterRecords
- GetWhatsAppProviderAdapterTimeline
- GetWhatsAppProviderAdapterEvidence
- GetWhatsAppProviderAdapterDecisionPackage
- GetWhatsAppProviderAdapterExceptions
- GetWhatsAppProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/whatsappprovideradapter/commands
GET  /api/v1/whatsappprovideradapter/{id}
GET  /api/v1/whatsappprovideradapter/{id}/timeline
GET  /api/v1/whatsappprovideradapter/{id}/evidence
GET  /api/v1/whatsappprovideradapter/{id}/decisions
POST /api/v1/whatsappprovideradapter/{id}/exceptions
POST /api/v1/whatsappprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappprovideradapter_aggregate	authoritative subject
whatsappprovideradapter_version	immutable submitted, approved, issued, or signed versions
whatsappprovideradapter_relationship	governed relationships
whatsappprovideradapter_decision	binding Decisions and rationale
whatsappprovideradapter_evidence_link	provenance and evidence relationships
whatsappprovideradapter_event_outbox	transactional Event publication
whatsappprovideradapter_idempotency	duplicate-write protection
whatsappprovideradapter_exception	exception, claim, dispute, or hold
whatsappprovideradapter_correction	correction and supersession lineage
whatsappprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-02-03
Subject	WhatsAppOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for WhatsApp.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppOperatingProfile {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppOperatingProfile
- ValidateWhatsAppOperatingProfile
- ApproveWhatsAppOperatingProfile
- ActivateWhatsAppOperatingProfile
- SuspendWhatsAppOperatingProfile
- CorrectWhatsAppOperatingProfile
- RetireWhatsAppOperatingProfile

Reference Events:

- WhatsAppOperatingProfileCreated
- WhatsAppOperatingProfileValidated
- WhatsAppOperatingProfileApproved
- WhatsAppOperatingProfileActivated
- WhatsAppOperatingProfileSuspended
- WhatsAppOperatingProfileCorrected
- WhatsAppOperatingProfileRetired

Reference queries:

- GetWhatsAppOperatingProfile
- ListWhatsAppOperatingProfileRecords
- GetWhatsAppOperatingProfileTimeline
- GetWhatsAppOperatingProfileEvidence
- GetWhatsAppOperatingProfileDecisionPackage
- GetWhatsAppOperatingProfileExceptions
- GetWhatsAppOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/whatsappoperatingprofile/commands
GET  /api/v1/whatsappoperatingprofile/{id}
GET  /api/v1/whatsappoperatingprofile/{id}/timeline
GET  /api/v1/whatsappoperatingprofile/{id}/evidence
GET  /api/v1/whatsappoperatingprofile/{id}/decisions
POST /api/v1/whatsappoperatingprofile/{id}/exceptions
POST /api/v1/whatsappoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappoperatingprofile_aggregate	authoritative subject
whatsappoperatingprofile_version	immutable submitted, approved, issued, or signed versions
whatsappoperatingprofile_relationship	governed relationships
whatsappoperatingprofile_decision	binding Decisions and rationale
whatsappoperatingprofile_evidence_link	provenance and evidence relationships
whatsappoperatingprofile_event_outbox	transactional Event publication
whatsappoperatingprofile_idempotency	duplicate-write protection
whatsappoperatingprofile_exception	exception, claim, dispute, or hold
whatsappoperatingprofile_correction	correction and supersession lineage
whatsappoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp Connector — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-03-00
Subject	WhatsAppConnectorChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp Connector
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for WhatsApp Connector.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp Connector**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppConnectorChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppConnectorChannel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppConnectorChannel
- ValidateWhatsAppConnectorChannel
- ApproveWhatsAppConnectorChannel
- ActivateWhatsAppConnectorChannel
- SuspendWhatsAppConnectorChannel
- CorrectWhatsAppConnectorChannel
- RetireWhatsAppConnectorChannel

Reference Events:

- WhatsAppConnectorChannelCreated
- WhatsAppConnectorChannelValidated
- WhatsAppConnectorChannelApproved
- WhatsAppConnectorChannelActivated
- WhatsAppConnectorChannelSuspended
- WhatsAppConnectorChannelCorrected
- WhatsAppConnectorChannelRetired

Reference queries:

- GetWhatsAppConnectorChannel
- ListWhatsAppConnectorChannelRecords
- GetWhatsAppConnectorChannelTimeline
- GetWhatsAppConnectorChannelEvidence
- GetWhatsAppConnectorChannelDecisionPackage
- GetWhatsAppConnectorChannelExceptions
- GetWhatsAppConnectorChannelAuditTrail

Reference API surface:

```
POST /api/v1/whatsappconnectorchannel/commands
GET  /api/v1/whatsappconnectorchannel/{id}
GET  /api/v1/whatsappconnectorchannel/{id}/timeline
GET  /api/v1/whatsappconnectorchannel/{id}/evidence
GET  /api/v1/whatsappconnectorchannel/{id}/decisions
POST /api/v1/whatsappconnectorchannel/{id}/exceptions
POST /api/v1/whatsappconnectorchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappconnectorchannel_aggregate	authoritative subject
whatsappconnectorchannel_version	immutable submitted, approved, issued, or signed versions
whatsappconnectorchannel_relationship	governed relationships
whatsappconnectorchannel_decision	binding Decisions and rationale
whatsappconnectorchannel_evidence_link	provenance and evidence relationships
whatsappconnectorchannel_event_outbox	transactional Event publication
whatsappconnectorchannel_idempotency	duplicate-write protection
whatsappconnectorchannel_exception	exception, claim, dispute, or hold
whatsappconnectorchannel_correction	correction and supersession lineage
whatsappconnectorchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp Connector — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-03-01
Subject	WhatsAppConnectorMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp Connector
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for WhatsApp Connector**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp Connector**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppConnectorMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppConnectorMessageModel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppConnectorMessageModel
- ValidateWhatsAppConnectorMessageModel
- ApproveWhatsAppConnectorMessageModel
- ActivateWhatsAppConnectorMessageModel
- SuspendWhatsAppConnectorMessageModel
- CorrectWhatsAppConnectorMessageModel
- RetireWhatsAppConnectorMessageModel

Reference Events:

- WhatsAppConnectorMessageModelCreated
- WhatsAppConnectorMessageModelValidated
- WhatsAppConnectorMessageModelApproved
- WhatsAppConnectorMessageModelActivated
- WhatsAppConnectorMessageModelSuspended
- WhatsAppConnectorMessageModelCorrected
- WhatsAppConnectorMessageModelRetired

Reference queries:

- GetWhatsAppConnectorMessageModel
- ListWhatsAppConnectorMessageModelRecords
- GetWhatsAppConnectorMessageModelTimeline
- GetWhatsAppConnectorMessageModelEvidence
- GetWhatsAppConnectorMessageModelDecisionPackage
- GetWhatsAppConnectorMessageModelExceptions
- GetWhatsAppConnectorMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/whatsappconnectormessagemodel/commands
GET  /api/v1/whatsappconnectormessagemodel/{id}
GET  /api/v1/whatsappconnectormessagemodel/{id}/timeline
GET  /api/v1/whatsappconnectormessagemodel/{id}/evidence
GET  /api/v1/whatsappconnectormessagemodel/{id}/decisions
POST /api/v1/whatsappconnectormessagemodel/{id}/exceptions
POST /api/v1/whatsappconnectormessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappconnectormessagemodel_aggregate	authoritative subject
whatsappconnectormessagemodel_version	immutable submitted, approved, issued, or signed versions
whatsappconnectormessagemodel_relationship	governed relationships
whatsappconnectormessagemodel_decision	binding Decisions and rationale
whatsappconnectormessagemodel_evidence_link	provenance and evidence relationships
whatsappconnectormessagemodel_event_outbox	transactional Event publication
whatsappconnectormessagemodel_idempotency	duplicate-write protection
whatsappconnectormessagemodel_exception	exception, claim, dispute, or hold
whatsappconnectormessagemodel_correction	correction and supersession lineage
whatsappconnectormessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp Connector — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-03-02
Subject	WhatsAppConnectorProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp Connector
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for WhatsApp Connector**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp Connector**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppConnectorProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppConnectorProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppConnectorProviderAdapter
- ValidateWhatsAppConnectorProviderAdapter
- ApproveWhatsAppConnectorProviderAdapter
- ActivateWhatsAppConnectorProviderAdapter
- SuspendWhatsAppConnectorProviderAdapter
- CorrectWhatsAppConnectorProviderAdapter
- RetireWhatsAppConnectorProviderAdapter

Reference Events:

- WhatsAppConnectorProviderAdapterCreated
- WhatsAppConnectorProviderAdapterValidated
- WhatsAppConnectorProviderAdapterApproved
- WhatsAppConnectorProviderAdapterActivated
- WhatsAppConnectorProviderAdapterSuspended
- WhatsAppConnectorProviderAdapterCorrected
- WhatsAppConnectorProviderAdapterRetired

Reference queries:

- GetWhatsAppConnectorProviderAdapter
- ListWhatsAppConnectorProviderAdapterRecords
- GetWhatsAppConnectorProviderAdapterTimeline
- GetWhatsAppConnectorProviderAdapterEvidence
- GetWhatsAppConnectorProviderAdapterDecisionPackage
- GetWhatsAppConnectorProviderAdapterExceptions
- GetWhatsAppConnectorProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/whatsappconnectorprovideradapter/commands
GET  /api/v1/whatsappconnectorprovideradapter/{id}
GET  /api/v1/whatsappconnectorprovideradapter/{id}/timeline
GET  /api/v1/whatsappconnectorprovideradapter/{id}/evidence
GET  /api/v1/whatsappconnectorprovideradapter/{id}/decisions
POST /api/v1/whatsappconnectorprovideradapter/{id}/exceptions
POST /api/v1/whatsappconnectorprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappconnectorprovideradapter_aggregate	authoritative subject
whatsappconnectorprovideradapter_version	immutable submitted, approved, issued, or signed versions
whatsappconnectorprovideradapter_relationship	governed relationships
whatsappconnectorprovideradapter_decision	binding Decisions and rationale
whatsappconnectorprovideradapter_evidence_link	provenance and evidence relationships
whatsappconnectorprovideradapter_event_outbox	transactional Event publication
whatsappconnectorprovideradapter_idempotency	duplicate-write protection
whatsappconnectorprovideradapter_exception	exception, claim, dispute, or hold
whatsappconnectorprovideradapter_correction	correction and supersession lineage
whatsappconnectorprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WhatsApp Connector — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-03-03
Subject	WhatsAppConnector0peratingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WhatsApp Connector
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for WhatsApp Connector.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WhatsApp Connector**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WhatsAppConnectorOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WhatsAppConnectorOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WhatsAppConnectorOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWhatsAppConnectorOperatingProfile
- ValidateWhatsAppConnectorOperatingProfile
- ApproveWhatsAppConnectorOperatingProfile
- ActivateWhatsAppConnectorOperatingProfile
- SuspendWhatsAppConnectorOperatingProfile
- CorrectWhatsAppConnectorOperatingProfile
- RetireWhatsAppConnectorOperatingProfile

Reference Events:

- WhatsAppConnectorOperatingProfileCreated
- WhatsAppConnectorOperatingProfileValidated
- WhatsAppConnectorOperatingProfileApproved
- WhatsAppConnectorOperatingProfileActivated
- WhatsAppConnectorOperatingProfileSuspended
- WhatsAppConnectorOperatingProfileCorrected
- WhatsAppConnectorOperatingProfileRetired

Reference queries:

- GetWhatsAppConnectorOperatingProfile
- ListWhatsAppConnectorOperatingProfileRecords
- GetWhatsAppConnectorOperatingProfileTimeline
- GetWhatsAppConnectorOperatingProfileEvidence
- GetWhatsAppConnectorOperatingProfileDecisionPackage
- GetWhatsAppConnectorOperatingProfileExceptions
- GetWhatsAppConnectorOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/whatsappconnectoroperatingprofile/commands
GET  /api/v1/whatsappconnectoroperatingprofile/{id}
GET  /api/v1/whatsappconnectoroperatingprofile/{id}/timeline
GET  /api/v1/whatsappconnectoroperatingprofile/{id}/evidence
GET  /api/v1/whatsappconnectoroperatingprofile/{id}/decisions
POST /api/v1/whatsappconnectoroperatingprofile/{id}/exceptions
POST /api/v1/whatsappconnectoroperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
whatsappconnectoroperatingprofile_aggregate	authoritative subject
whatsappconnectoroperatingprofile_version	immutable submitted, approved, issued, or signed versions
whatsappconnectoroperatingprofile_relationship	governed relationships
whatsappconnectoroperatingprofile_decision	binding Decisions and rationale
whatsappconnectoroperatingprofile_evidence_link	provenance and evidence relationships
whatsappconnectoroperatingprofile_event_outbox	transactional Event publication
whatsappconnectoroperatingprofile_idempotency	duplicate-write protection
whatsappconnectoroperatingprofile_exception	exception, claim, dispute, or hold
whatsappconnectoroperatingprofile_correction	correction and supersession lineage
whatsappconnectoroperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Instagram — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-04-00
Subject	InstagramChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Instagram
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Instagram.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Instagram**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
InstagramChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
InstagramChannel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateInstagramChannel
- ValidateInstagramChannel
- ApproveInstagramChannel
- ActivateInstagramChannel
- SuspendInstagramChannel
- CorrectInstagramChannel
- RetireInstagramChannel

Reference Events:

- InstagramChannelCreated
- InstagramChannelValidated
- InstagramChannelApproved
- InstagramChannelActivated
- InstagramChannelSuspended
- InstagramChannelCorrected
- InstagramChannelRetired

Reference queries:

- GetInstagramChannel
- ListInstagramChannelRecords
- GetInstagramChannelTimeline
- GetInstagramChannelEvidence
- GetInstagramChannelDecisionPackage
- GetInstagramChannelExceptions
- GetInstagramChannelAuditTrail

Reference API surface:

```
POST /api/v1/instagramchannel/commands
GET  /api/v1/instagramchannel/{id}
GET  /api/v1/instagramchannel/{id}/timeline
GET  /api/v1/instagramchannel/{id}/evidence
GET  /api/v1/instagramchannel/{id}/decisions
POST /api/v1/instagramchannel/{id}/exceptions
POST /api/v1/instagramchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
instagramchannel_aggregate	authoritative subject
instagramchannel_version	immutable submitted, approved, issued, or signed versions
instagramchannel_relationship	governed relationships
instagramchannel_decision	binding Decisions and rationale
instagramchannel_evidence_link	provenance and evidence relationships
instagramchannel_event_outbox	transactional Event publication
instagramchannel_idempotency	duplicate-write protection
instagramchannel_exception	exception, claim, dispute, or hold
instagramchannel_correction	correction and supersession lineage
instagramchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Instagram — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-04-01
Subject	InstagramMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Instagram
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Instagram.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Instagram**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
InstagramMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
InstagramMessageModel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateInstagramMessageModel
- ValidateInstagramMessageModel
- ApproveInstagramMessageModel
- ActivateInstagramMessageModel
- SuspendInstagramMessageModel
- CorrectInstagramMessageModel
- RetireInstagramMessageModel

Reference Events:

- InstagramMessageModelCreated
- InstagramMessageModelValidated
- InstagramMessageModelApproved
- InstagramMessageModelActivated
- InstagramMessageModelSuspended
- InstagramMessageModelCorrected
- InstagramMessageModelRetired

Reference queries:

- GetInstagramMessageModel
- ListInstagramMessageModelRecords
- GetInstagramMessageModelTimeline
- GetInstagramMessageModelEvidence
- GetInstagramMessageModelDecisionPackage
- GetInstagramMessageModelExceptions
- GetInstagramMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/instagrammessagemodel/commands
GET  /api/v1/instagrammessagemodel/{id}
GET  /api/v1/instagrammessagemodel/{id}/timeline
GET  /api/v1/instagrammessagemodel/{id}/evidence
GET  /api/v1/instagrammessagemodel/{id}/decisions
POST /api/v1/instagrammessagemodel/{id}/exceptions
POST /api/v1/instagrammessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
instagrammessagemodel_aggregate	authoritative subject
instagrammessagemodel_version	immutable submitted, approved, issued, or signed versions
instagrammessagemodel_relationship	governed relationships
instagrammessagemodel_decision	binding Decisions and rationale
instagrammessagemodel_evidence_link	provenance and evidence relationships
instagrammessagemodel_event_outbox	transactional Event publication
instagrammessagemodel_idempotency	duplicate-write protection
instagrammessagemodel_exception	exception, claim, dispute, or hold
instagrammessagemodel_correction	correction and supersession lineage
instagrammessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Instagram — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-04-02
Subject	InstagramProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Instagram
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Instagram.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Instagram**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
InstagramProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
InstagramProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateInstagramProviderAdapter
- ValidateInstagramProviderAdapter
- ApproveInstagramProviderAdapter
- ActivateInstagramProviderAdapter
- SuspendInstagramProviderAdapter
- CorrectInstagramProviderAdapter
- RetireInstagramProviderAdapter

Reference Events:

- InstagramProviderAdapterCreated
- InstagramProviderAdapterValidated
- InstagramProviderAdapterApproved
- InstagramProviderAdapterActivated
- InstagramProviderAdapterSuspended
- InstagramProviderAdapterCorrected
- InstagramProviderAdapterRetired

Reference queries:

- GetInstagramProviderAdapter
- ListInstagramProviderAdapterRecords
- GetInstagramProviderAdapterTimeline
- GetInstagramProviderAdapterEvidence
- GetInstagramProviderAdapterDecisionPackage
- GetInstagramProviderAdapterExceptions
- GetInstagramProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/instagramprovideradapter/commands
GET  /api/v1/instagramprovideradapter/{id}
GET  /api/v1/instagramprovideradapter/{id}/timeline
GET  /api/v1/instagramprovideradapter/{id}/evidence
GET  /api/v1/instagramprovideradapter/{id}/decisions
POST /api/v1/instagramprovideradapter/{id}/exceptions
POST /api/v1/instagramprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
instagramprovideradapter_aggregate	authoritative subject
instagramprovideradapter_version	immutable submitted, approved, issued, or signed versions
instagramprovideradapter_relationship	governed relationships
instagramprovideradapter_decision	binding Decisions and rationale
instagramprovideradapter_evidence_link	provenance and evidence relationships
instagramprovideradapter_event_outbox	transactional Event publication
instagramprovideradapter_idempotency	duplicate-write protection
instagramprovideradapter_exception	exception, claim, dispute, or hold
instagramprovideradapter_correction	correction and supersession lineage
instagramprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Instagram — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-04-03
Subject	InstagramOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Instagram
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Instagram.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Instagram**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
InstagramOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
InstagramOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
InstagramOperatingProfile {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateInstagramOperatingProfile
- ValidateInstagramOperatingProfile
- ApproveInstagramOperatingProfile
- ActivateInstagramOperatingProfile
- SuspendInstagramOperatingProfile
- CorrectInstagramOperatingProfile
- RetireInstagramOperatingProfile

Reference Events:

- InstagramOperatingProfileCreated
- InstagramOperatingProfileValidated
- InstagramOperatingProfileApproved
- InstagramOperatingProfileActivated
- InstagramOperatingProfileSuspended
- InstagramOperatingProfileCorrected
- InstagramOperatingProfileRetired

Reference queries:

- GetInstagramOperatingProfile
- ListInstagramOperatingProfileRecords
- GetInstagramOperatingProfileTimeline
- GetInstagramOperatingProfileEvidence
- GetInstagramOperatingProfileDecisionPackage
- GetInstagramOperatingProfileExceptions
- GetInstagramOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/instagramoperatingprofile/commands
GET  /api/v1/instagramoperatingprofile/{id}
GET  /api/v1/instagramoperatingprofile/{id}/timeline
GET  /api/v1/instagramoperatingprofile/{id}/evidence
GET  /api/v1/instagramoperatingprofile/{id}/decisions
POST /api/v1/instagramoperatingprofile/{id}/exceptions
POST /api/v1/instagramoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
instagramoperatingprofile_aggregate	authoritative subject
instagramoperatingprofile_version	immutable submitted, approved, issued, or signed versions
instagramoperatingprofile_relationship	governed relationships
instagramoperatingprofile_decision	binding Decisions and rationale
instagramoperatingprofile_evidence_link	provenance and evidence relationships
instagramoperatingprofile_event_outbox	transactional Event publication
instagramoperatingprofile_idempotency	duplicate-write protection
instagramoperatingprofile_exception	exception, claim, dispute, or hold
instagramoperatingprofile_correction	correction and supersession lineage
instagramoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Email — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-05-00
Subject	EmailChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Email
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Email**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Email**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EmailChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EmailChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEmailChannel
- ValidateEmailChannel
- ApproveEmailChannel
- ActivateEmailChannel
- SuspendEmailChannel
- CorrectEmailChannel
- RetireEmailChannel

Reference Events:

- EmailChannelCreated
- EmailChannelValidated
- EmailChannelApproved
- EmailChannelActivated
- EmailChannelSuspended
- EmailChannelCorrected
- EmailChannelRetired

Reference queries:

- GetEmailChannel
- ListEmailChannelRecords
- GetEmailChannelTimeline
- GetEmailChannelEvidence
- GetEmailChannelDecisionPackage
- GetEmailChannelExceptions
- GetEmailChannelAuditTrail

Reference API surface:

```

POST /api/v1/emailchannel/commands
GET  /api/v1/emailchannel/{id}
GET  /api/v1/emailchannel/{id}/timeline
GET  /api/v1/emailchannel/{id}/evidence
GET  /api/v1/emailchannel/{id}/decisions
POST /api/v1/emailchannel/{id}/exceptions
POST /api/v1/emailchannel/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Policy and evidence gates pass?]
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
emailchannel_aggregate	authoritative subject
emailchannel_version	immutable submitted, approved, issued, or signed versions
emailchannel_relationship	governed relationships
emailchannel_decision	binding Decisions and rationale
emailchannel_evidence_link	provenance and evidence relationships
emailchannel_event_outbox	transactional Event publication
emailchannel_idempotency	duplicate-write protection
emailchannel_exception	exception, claim, dispute, or hold
emailchannel_correction	correction and supersession lineage
emailchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history

- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Email — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-05-01
Subject	EmailMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Email
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Email**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Email**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EmailMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EmailMessageModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEmailMessageModel
- ValidateEmailMessageModel
- ApproveEmailMessageModel
- ActivateEmailMessageModel
- SuspendEmailMessageModel
- CorrectEmailMessageModel
- RetireEmailMessageModel

Reference Events:

- EmailMessageModelCreated
- EmailMessageModelValidated
- EmailMessageModelApproved
- EmailMessageModelActivated
- EmailMessageModelSuspended
- EmailMessageModelCorrected
- EmailMessageModelRetired

Reference queries:

- GetEmailMessageModel
- ListEmailMessageModelRecords
- GetEmailMessageModelTimeline
- GetEmailMessageModelEvidence
- GetEmailMessageModelDecisionPackage
- GetEmailMessageModelExceptions
- GetEmailMessageModelAuditTrail

Reference API surface:

```

POST /api/v1/emailmessagemodel/commands
GET  /api/v1/emailmessagemodel/{id}
GET  /api/v1/emailmessagemodel/{id}/timeline
GET  /api/v1/emailmessagemodel/{id}/evidence
GET  /api/v1/emailmessagemodel/{id}/decisions
POST /api/v1/emailmessagemodel/{id}/exceptions
POST /api/v1/emailmessagemodel/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Policy and evidence gates pass?]
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
emailmessagemodel_aggregate	authoritative subject
emailmessagemodel_version	immutable submitted, approved, issued, or signed versions
emailmessagemodel_relationship	governed relationships
emailmessagemodel_decision	binding Decisions and rationale
emailmessagemodel_evidence_link	provenance and evidence relationships
emailmessagemodel_event_outbox	transactional Event publication
emailmessagemodel_idempotency	duplicate-write protection
emailmessagemodel_exception	exception, claim, dispute, or hold
emailmessagemodel_correction	correction and supersession lineage
emailmessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history

- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Email — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-05-02
Subject	EmailProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Email
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Email**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Email**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EmailProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EmailProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEmailProviderAdapter
- ValidateEmailProviderAdapter
- ApproveEmailProviderAdapter
- ActivateEmailProviderAdapter
- SuspendEmailProviderAdapter
- CorrectEmailProviderAdapter
- RetireEmailProviderAdapter

Reference Events:

- EmailProviderAdapterCreated
- EmailProviderAdapterValidated
- EmailProviderAdapterApproved
- EmailProviderAdapterActivated
- EmailProviderAdapterSuspended
- EmailProviderAdapterCorrected
- EmailProviderAdapterRetired

Reference queries:

- GetEmailProviderAdapter
- ListEmailProviderAdapterRecords
- GetEmailProviderAdapterTimeline
- GetEmailProviderAdapterEvidence
- GetEmailProviderAdapterDecisionPackage
- GetEmailProviderAdapterExceptions
- GetEmailProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/emailprovideradapter/commands
GET  /api/v1/emailprovideradapter/{id}
GET  /api/v1/emailprovideradapter/{id}/timeline
GET  /api/v1/emailprovideradapter/{id}/evidence
GET  /api/v1/emailprovideradapter/{id}/decisions
POST /api/v1/emailprovideradapter/{id}/exceptions
POST /api/v1/emailprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
emailprovideradapter_aggregate	authoritative subject
emailprovideradapter_version	immutable submitted, approved, issued, or signed versions
emailprovideradapter_relationship	governed relationships
emailprovideradapter_decision	binding Decisions and rationale
emailprovideradapter_evidence_link	provenance and evidence relationships
emailprovideradapter_event_outbox	transactional Event publication
emailprovideradapter_idempotency	duplicate-write protection
emailprovideradapter_exception	exception, claim, dispute, or hold
emailprovideradapter_correction	correction and supersession lineage
emailprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Email — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-05-03
Subject	Email0peratingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Email
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Email.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Email**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EmailOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EmailOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EmailOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEmailOperatingProfile
- ValidateEmailOperatingProfile
- ApproveEmailOperatingProfile
- ActivateEmailOperatingProfile
- SuspendEmailOperatingProfile
- CorrectEmailOperatingProfile
- RetireEmailOperatingProfile

Reference Events:

- EmailOperatingProfileCreated
- EmailOperatingProfileValidated
- EmailOperatingProfileApproved
- EmailOperatingProfileActivated
- EmailOperatingProfileSuspended
- EmailOperatingProfileCorrected
- EmailOperatingProfileRetired

Reference queries:

- GetEmailOperatingProfile
- ListEmailOperatingProfileRecords
- GetEmailOperatingProfileTimeline
- GetEmailOperatingProfileEvidence
- GetEmailOperatingProfileDecisionPackage
- GetEmailOperatingProfileExceptions
- GetEmailOperatingProfileAuditTrail

Reference API surface:

```

POST /api/v1/emailoperatingprofile/commands
GET  /api/v1/emailoperatingprofile/{id}
GET  /api/v1/emailoperatingprofile/{id}/timeline
GET  /api/v1/emailoperatingprofile/{id}/evidence
GET  /api/v1/emailoperatingprofile/{id}/decisions
POST /api/v1/emailoperatingprofile/{id}/exceptions
POST /api/v1/emailoperatingprofile/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
emailoperatingprofile_aggregate	authoritative subject
emailoperatingprofile_version	immutable submitted, approved, issued, or signed versions
emailoperatingprofile_relationship	governed relationships
emailoperatingprofile_decision	binding Decisions and rationale
emailoperatingprofile_evidence_link	provenance and evidence relationships
emailoperatingprofile_event_outbox	transactional Event publication
emailoperatingprofile_idempotency	duplicate-write protection
emailoperatingprofile_exception	exception, claim, dispute, or hold
emailoperatingprofile_correction	correction and supersession lineage
emailoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history

- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
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duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WebSocket and Realtime Messaging — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-06-00
Subject	WebSocketandRealtimeMessagingChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WebSocket and Realtime Messaging
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for WebSocket and Realtime Messaging.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WebSocket and Realtime Messaging.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WebSocketandRealtimeMessagingChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WebSocketandRealtimeMessagingChannel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWebSocketandRealtimeMessagingChannel
- ValidateWebSocketandRealtimeMessagingChannel
- ApproveWebSocketandRealtimeMessagingChannel
- ActivateWebSocketandRealtimeMessagingChannel
- SuspendWebSocketandRealtimeMessagingChannel
- CorrectWebSocketandRealtimeMessagingChannel
- RetireWebSocketandRealtimeMessagingChannel

Reference Events:

- WebSocketandRealtimeMessagingChannelCreated
- WebSocketandRealtimeMessagingChannelValidated
- WebSocketandRealtimeMessagingChannelApproved
- WebSocketandRealtimeMessagingChannelActivated
- WebSocketandRealtimeMessagingChannelSuspended
- WebSocketandRealtimeMessagingChannelCorrected
- WebSocketandRealtimeMessagingChannelRetired

Reference queries:

- GetWebSocketandRealtimeMessagingChannel
- ListWebSocketandRealtimeMessagingChannelRecords
- GetWebSocketandRealtimeMessagingChannelTimeline
- GetWebSocketandRealtimeMessagingChannelEvidence
- GetWebSocketandRealtimeMessagingChannelDecisionPackage
- GetWebSocketandRealtimeMessagingChannelExceptions
- GetWebSocketandRealtimeMessagingChannelAuditTrail

Reference API surface:

```
POST /api/v1/websocketandrealtimemessagingchannel/commands
GET  /api/v1/websocketandrealtimemessagingchannel/{id}
GET  /api/v1/websocketandrealtimemessagingchannel/{id}/timeline
GET  /api/v1/websocketandrealtimemessagingchannel/{id}/evidence
GET  /api/v1/websocketandrealtimemessagingchannel/{id}/decisions
POST /api/v1/websocketandrealtimemessagingchannel/{id}/exceptions
POST /api/v1/websocketandrealtimemessagingchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
websocketandrealtimeessagingchannel_aggregate	authoritative subject
websocketandrealtimeessagingchannel_version	immutable submitted, approved, issued, or signed versions
websocketandrealtimeessagingchannel_relationship	governed relationships
websocketandrealtimeessagingchannel_decision	binding Decisions and rationale
websocketandrealtimeessagingchannel_evidence_link	provenance and evidence relationships
websocketandrealtimeessagingchannel_event_outbox	transactional Event publication
websocketandrealtimeessagingchannel_idempotency	duplicate-write protection
websocketandrealtimeessagingchannel_exception	exception, claim, dispute, or hold
websocketandrealtimeessagingchannel_correction	correction and supersession lineage
websocketandrealtimeessagingchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WebSocket and Realtime Messaging — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-06-01
Subject	WebSocketandRealtimeMessagingMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WebSocket and Realtime Messaging
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for WebSocket and Realtime Messaging**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WebSocket and Realtime Messaging**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WebSocketandRealtimeMessagingMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WebSocketandRealtimeMessagingMessageModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWebSocketandRealtimeMessagingMessageModel
- ValidateWebSocketandRealtimeMessagingMessageModel
- ApproveWebSocketandRealtimeMessagingMessageModel
- ActivateWebSocketandRealtimeMessagingMessageModel
- SuspendWebSocketandRealtimeMessagingMessageModel
- CorrectWebSocketandRealtimeMessagingMessageModel
- RetireWebSocketandRealtimeMessagingMessageModel

Reference Events:

- WebSocketandRealtimeMessagingMessageModelCreated
- WebSocketandRealtimeMessagingMessageModelValidated
- WebSocketandRealtimeMessagingMessageModelApproved
- WebSocketandRealtimeMessagingMessageModelActivated
- WebSocketandRealtimeMessagingMessageModelSuspended
- WebSocketandRealtimeMessagingMessageModelCorrected
- WebSocketandRealtimeMessagingMessageModelRetired

Reference queries:

- GetWebSocketandRealtimeMessagingMessageModel
- ListWebSocketandRealtimeMessagingMessageModelRecords
- GetWebSocketandRealtimeMessagingMessageModelTimeline
- GetWebSocketandRealtimeMessagingMessageModelEvidence
- GetWebSocketandRealtimeMessagingMessageModelDecisionPackage
- GetWebSocketandRealtimeMessagingMessageModelExceptions
- GetWebSocketandRealtimeMessagingMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/websocketandrealtimeessagingmessagemodel/commands
GET  /api/v1/websocketandrealtimeessagingmessagemodel/{id}
GET  /api/v1/websocketandrealtimeessagingmessagemodel/{id}/timeline
GET  /api/v1/websocketandrealtimeessagingmessagemodel/{id}/evidence
GET  /api/v1/websocketandrealtimeessagingmessagemodel/{id}/decisions
POST /api/v1/websocketandrealtimeessagingmessagemodel/{id}/exceptions
POST /api/v1/websocketandrealtimeessagingmessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
websocketandrealtimeessagingmessagemodel_ aggregate	authoritative subject
websocketandrealtimeessagingmessagemodel_ version	immutable submitted, approved, issued, or signed versions
websocketandrealtimeessagingmessagemodel_ relationship	governed relationships
websocketandrealtimeessagingmessagemodel_ decision	binding Decisions and rationale
websocketandrealtimeessagingmessagemodel_ evidence_link	provenance and evidence relationships
websocketandrealtimeessagingmessagemodel_ event_ outbox	transactional Event publication
websocketandrealtimeessagingmessagemodel_ idempotency	duplicate-write protection
websocketandrealtimeessagingmessagemodel_ exception	exception, claim, dispute, or hold
websocketandrealtimeessagingmessagemodel_ correction	correction and supersession lineage
websocketandrealtimeessagingmessagemodel_ audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.

- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.

Status	Meaning
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

WebSocket and Realtime Messaging — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-06-02
Subject	WebSocketandRealtimeMessagingProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WebSocket and Realtime Messaging
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for WebSocket and Realtime Messaging**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WebSocket and Realtime Messaging**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WebSocketandRealtimeMessagingProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingProviderAdapterEvidence	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WebSocketandRealtimeMessagingProviderAdapter {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWebSocketandRealtimeMessagingProviderAdapter
- ValidateWebSocketandRealtimeMessagingProviderAdapter
- ApproveWebSocketandRealtimeMessagingProviderAdapter
- ActivateWebSocketandRealtimeMessagingProviderAdapter
- SuspendWebSocketandRealtimeMessagingProviderAdapter
- CorrectWebSocketandRealtimeMessagingProviderAdapter
- RetireWebSocketandRealtimeMessagingProviderAdapter

Reference Events:

- WebSocketandRealtimeMessagingProviderAdapterCreated
- WebSocketandRealtimeMessagingProviderAdapterValidated
- WebSocketandRealtimeMessagingProviderAdapterApproved
- WebSocketandRealtimeMessagingProviderAdapterActivated
- WebSocketandRealtimeMessagingProviderAdapterSuspended
- WebSocketandRealtimeMessagingProviderAdapterCorrected
- WebSocketandRealtimeMessagingProviderAdapterRetired

Reference queries:

- GetWebSocketandRealtimeMessagingProviderAdapter
- ListWebSocketandRealtimeMessagingProviderAdapterRecords
- GetWebSocketandRealtimeMessagingProviderAdapterTimeline
- GetWebSocketandRealtimeMessagingProviderAdapterEvidence
- GetWebSocketandRealtimeMessagingProviderAdapterDecisionPackage
- GetWebSocketandRealtimeMessagingProviderAdapterExceptions
- GetWebSocketandRealtimeMessagingProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/websocketandrealtimemessagingprovideradapter/commands
GET  /api/v1/websocketandrealtimemessagingprovideradapter/{id}
GET  /api/v1/websocketandrealtimemessagingprovideradapter/{id}/timeline
GET  /api/v1/websocketandrealtimemessagingprovideradapter/{id}/evidence
GET  /api/v1/websocketandrealtimemessagingprovideradapter/{id}/decisions
POST /api/v1/websocketandrealtimemessagingprovideradapter/{id}/exceptions
POST /api/v1/websocketandrealtimemessagingprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
websocketandrealtimeessagingprovideradapter_aggregate	authoritative subject
websocketandrealtimeessagingprovideradapter_version	immutable submitted, approved, issued, or signed versions
websocketandrealtimeessagingprovideradapter_relationship	governed relationships
websocketandrealtimeessagingprovideradapter_decision	binding Decisions and rationale
websocketandrealtimeessagingprovideradapter_evidence_link	provenance and evidence relationships
websocketandrealtimeessagingprovideradapter_event_outbox	transactional Event publication
websocketandrealtimeessagingprovideradapter_idempotency	duplicate-write protection
websocketandrealtimeessagingprovideradapter_exception	exception, claim, dispute, or hold
websocketandrealtimeessagingprovideradapter_correction	correction and supersession lineage
websocketandrealtimeessagingprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

WebSocket and Realtime Messaging — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-06-03
Subject	WebSocketandRealtimeMessagingOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / WebSocket and Realtime Messaging
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for WebSocket and Realtime Messaging.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / WebSocket and Realtime Messaging.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WebSocketandRealtimeMessagingOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingOperatingProfileEvidence	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WebSocketandRealtimeMessagingOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WebSocketandRealtimeMessagingOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWebSocketandRealtimeMessagingOperatingProfile
- ValidateWebSocketandRealtimeMessagingOperatingProfile
- ApproveWebSocketandRealtimeMessagingOperatingProfile
- ActivateWebSocketandRealtimeMessagingOperatingProfile
- SuspendWebSocketandRealtimeMessagingOperatingProfile
- CorrectWebSocketandRealtimeMessagingOperatingProfile
- RetireWebSocketandRealtimeMessagingOperatingProfile

Reference Events:

- WebSocketandRealtimeMessagingOperatingProfileCreated
- WebSocketandRealtimeMessagingOperatingProfileValidated
- WebSocketandRealtimeMessagingOperatingProfileApproved
- WebSocketandRealtimeMessagingOperatingProfileActivated
- WebSocketandRealtimeMessagingOperatingProfileSuspended
- WebSocketandRealtimeMessagingOperatingProfileCorrected
- WebSocketandRealtimeMessagingOperatingProfileRetired

Reference queries:

- GetWebSocketandRealtimeMessagingOperatingProfile
- ListWebSocketandRealtimeMessagingOperatingProfileRecords
- GetWebSocketandRealtimeMessagingOperatingProfileTimeline
- GetWebSocketandRealtimeMessagingOperatingProfileEvidence
- GetWebSocketandRealtimeMessagingOperatingProfileDecisionPackage
- GetWebSocketandRealtimeMessagingOperatingProfileExceptions
- GetWebSocketandRealtimeMessagingOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/websocketandrealtimemessagingoperatingprofile/commands
GET  /api/v1/websocketandrealtimemessagingoperatingprofile/{id}
GET  /api/v1/websocketandrealtimemessagingoperatingprofile/{id}/timeline
GET  /api/v1/websocketandrealtimemessagingoperatingprofile/{id}/evidence
GET  /api/v1/websocketandrealtimemessagingoperatingprofile/{id}/decisions
POST /api/v1/websocketandrealtimemessagingoperatingprofile/{id}/exceptions
POST /api/v1/websocketandrealtimemessagingoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
websocketandrealtimeessagingoperatingprofile_ aggregate	authoritative subject
websocketandrealtimeessagingoperatingprofile_ version	immutable submitted, approved, issued, or signed versions
websocketandrealtimeessagingoperatingprofile_ relationship	governed relationships
websocketandrealtimeessagingoperatingprofile_ decision	binding Decisions and rationale
websocketandrealtimeessagingoperatingprofile_ evidence_link	provenance and evidence relationships
websocketandrealtimeessagingoperatingprofile_ event_outbox	transactional Event publication
websocketandrealtimeessagingoperatingprofile_ idempotency	duplicate-write protection
websocketandrealtimeessagingoperatingprofile_ exception	exception, claim, dispute, or hold
websocketandrealtimeessagingoperatingprofile_ correction	correction and supersession lineage
websocketandrealtimeessagingoperatingprofile_ audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Conversation Ownership — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-07-00
Subject	ConversationOwnershipChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Conversation Ownership
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Conversation Ownership**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Conversation Ownership**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConversationOwnershipChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConversationOwnershipChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConversationOwnershipChannel
- ValidateConversationOwnershipChannel
- ApproveConversationOwnershipChannel
- ActivateConversationOwnershipChannel
- SuspendConversationOwnershipChannel
- CorrectConversationOwnershipChannel
- RetireConversationOwnershipChannel

Reference Events:

- ConversationOwnershipChannelCreated
- ConversationOwnershipChannelValidated
- ConversationOwnershipChannelApproved
- ConversationOwnershipChannelActivated
- ConversationOwnershipChannelSuspended
- ConversationOwnershipChannelCorrected
- ConversationOwnershipChannelRetired

Reference queries:

- GetConversationOwnershipChannel
- ListConversationOwnershipChannelRecords
- GetConversationOwnershipChannelTimeline
- GetConversationOwnershipChannelEvidence
- GetConversationOwnershipChannelDecisionPackage
- GetConversationOwnershipChannelExceptions
- GetConversationOwnershipChannelAuditTrail

Reference API surface:

```
POST /api/v1/conversationownershipchannel/commands
GET  /api/v1/conversationownershipchannel/{id}
GET  /api/v1/conversationownershipchannel/{id}/timeline
GET  /api/v1/conversationownershipchannel/{id}/evidence
GET  /api/v1/conversationownershipchannel/{id}/decisions
POST /api/v1/conversationownershipchannel/{id}/exceptions
POST /api/v1/conversationownershipchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
conversationownershipchannel_aggregate	authoritative subject
conversationownershipchannel_version	immutable submitted, approved, issued, or signed versions
conversationownershipchannel_relationship	governed relationships
conversationownershipchannel_decision	binding Decisions and rationale
conversationownershipchannel_evidence_link	provenance and evidence relationships
conversationownershipchannel_event_outbox	transactional Event publication
conversationownershipchannel_idempotency	duplicate-write protection
conversationownershipchannel_exception	exception, claim, dispute, or hold
conversationownershipchannel_correction	correction and supersession lineage
conversationownershipchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Conversation Ownership — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-07-01
Subject	ConversationOwnershipMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Conversation Ownership
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Conversation Ownership**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Conversation Ownership**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConversationOwnershipMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConversationOwnershipMessageModel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConversationOwnershipMessageModel
- ValidateConversationOwnershipMessageModel
- ApproveConversationOwnershipMessageModel
- ActivateConversationOwnershipMessageModel
- SuspendConversationOwnershipMessageModel
- CorrectConversationOwnershipMessageModel
- RetireConversationOwnershipMessageModel

Reference Events:

- ConversationOwnershipMessageModelCreated
- ConversationOwnershipMessageModelValidated
- ConversationOwnershipMessageModelApproved
- ConversationOwnershipMessageModelActivated
- ConversationOwnershipMessageModelSuspended
- ConversationOwnershipMessageModelCorrected
- ConversationOwnershipMessageModelRetired

Reference queries:

- GetConversationOwnershipMessageModel
- ListConversationOwnershipMessageModelRecords
- GetConversationOwnershipMessageModelTimeline
- GetConversationOwnershipMessageModelEvidence
- GetConversationOwnershipMessageModelDecisionPackage
- GetConversationOwnershipMessageModelExceptions
- GetConversationOwnershipMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/conversationownershipmessagemodel/commands
GET  /api/v1/conversationownershipmessagemodel/{id}
GET  /api/v1/conversationownershipmessagemodel/{id}/timeline
GET  /api/v1/conversationownershipmessagemodel/{id}/evidence
GET  /api/v1/conversationownershipmessagemodel/{id}/decisions
POST /api/v1/conversationownershipmessagemodel/{id}/exceptions
POST /api/v1/conversationownershipmessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
conversationownershipmessagemodel_aggregate	authoritative subject
conversationownershipmessagemodel_version	immutable submitted, approved, issued, or signed versions
conversationownershipmessagemodel_relationship	governed relationships
conversationownershipmessagemodel_decision	binding Decisions and rationale
conversationownershipmessagemodel_evidence_link	provenance and evidence relationships
conversationownershipmessagemodel_event_outbox	transactional Event publication
conversationownershipmessagemodel_idempotency	duplicate-write protection
conversationownershipmessagemodel_exception	exception, claim, dispute, or hold
conversationownershipmessagemodel_correction	correction and supersession lineage
conversationownershipmessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Conversation Ownership — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-07-02
Subject	ConversationOwnershipProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Conversation Ownership
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Conversation Ownership**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Conversation Ownership**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConversationOwnershipProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConversationOwnershipProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConversationOwnershipProviderAdapter
- ValidateConversationOwnershipProviderAdapter
- ApproveConversationOwnershipProviderAdapter
- ActivateConversationOwnershipProviderAdapter
- SuspendConversationOwnershipProviderAdapter
- CorrectConversationOwnershipProviderAdapter
- RetireConversationOwnershipProviderAdapter

Reference Events:

- ConversationOwnershipProviderAdapterCreated
- ConversationOwnershipProviderAdapterValidated
- ConversationOwnershipProviderAdapterApproved
- ConversationOwnershipProviderAdapterActivated
- ConversationOwnershipProviderAdapterSuspended
- ConversationOwnershipProviderAdapterCorrected
- ConversationOwnershipProviderAdapterRetired

Reference queries:

- GetConversationOwnershipProviderAdapter
- ListConversationOwnershipProviderAdapterRecords
- GetConversationOwnershipProviderAdapterTimeline
- GetConversationOwnershipProviderAdapterEvidence
- GetConversationOwnershipProviderAdapterDecisionPackage
- GetConversationOwnershipProviderAdapterExceptions
- GetConversationOwnershipProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/conversationownershipprovideradapter/commands
GET  /api/v1/conversationownershipprovideradapter/{id}
GET  /api/v1/conversationownershipprovideradapter/{id}/timeline
GET  /api/v1/conversationownershipprovideradapter/{id}/evidence
GET  /api/v1/conversationownershipprovideradapter/{id}/decisions
POST /api/v1/conversationownershipprovideradapter/{id}/exceptions
POST /api/v1/conversationownershipprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
conversationownershipprovideradapter_aggregate	authoritative subject
conversationownershipprovideradapter_version	immutable submitted, approved, issued, or signed versions
conversationownershipprovideradapter_relationship	governed relationships
conversationownershipprovideradapter_decision	binding Decisions and rationale
conversationownershipprovideradapter_evidence_link	provenance and evidence relationships
conversationownershipprovideradapter_event_outbox	transactional Event publication
conversationownershipprovideradapter_idempotency	duplicate-write protection
conversationownershipprovideradapter_exception	exception, claim, dispute, or hold
conversationownershipprovideradapter_correction	correction and supersession lineage
conversationownershipprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Conversation Ownership — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-07-03
Subject	ConversationOwnershipOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Conversation Ownership
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Conversation Ownership.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Conversation Ownership**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConversationOwnershipOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConversationOwnershipOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConversationOwnershipOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConversationOwnershipOperatingProfile
- ValidateConversationOwnershipOperatingProfile
- ApproveConversationOwnershipOperatingProfile
- ActivateConversationOwnershipOperatingProfile
- SuspendConversationOwnershipOperatingProfile
- CorrectConversationOwnershipOperatingProfile
- RetireConversationOwnershipOperatingProfile

Reference Events:

- ConversationOwnershipOperatingProfileCreated
- ConversationOwnershipOperatingProfileValidated
- ConversationOwnershipOperatingProfileApproved
- ConversationOwnershipOperatingProfileActivated
- ConversationOwnershipOperatingProfileSuspended
- ConversationOwnershipOperatingProfileCorrected
- ConversationOwnershipOperatingProfileRetired

Reference queries:

- GetConversationOwnershipOperatingProfile
- ListConversationOwnershipOperatingProfileRecords
- GetConversationOwnershipOperatingProfileTimeline
- GetConversationOwnershipOperatingProfileEvidence
- GetConversationOwnershipOperatingProfileDecisionPackage
- GetConversationOwnershipOperatingProfileExceptions
- GetConversationOwnershipOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/conversationownershipoperatingprofile/commands
GET  /api/v1/conversationownershipoperatingprofile/{id}
GET  /api/v1/conversationownershipoperatingprofile/{id}/timeline
GET  /api/v1/conversationownershipoperatingprofile/{id}/evidence
GET  /api/v1/conversationownershipoperatingprofile/{id}/decisions
POST /api/v1/conversationownershipoperatingprofile/{id}/exceptions
POST /api/v1/conversationownershipoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
conversationownershipoperatingprofile_aggregate	authoritative subject
conversationownershipoperatingprofile_version	immutable submitted, approved, issued, or signed versions
conversationownershipoperatingprofile_relationship	governed relationships
conversationownershipoperatingprofile_decision	binding Decisions and rationale
conversationownershipoperatingprofile_evidence_link	provenance and evidence relationships
conversationownershipoperatingprofile_event_outbox	transactional Event publication
conversationownershipoperatingprofile_idempotency	duplicate-write protection
conversationownershipoperatingprofile_exception	exception, claim, dispute, or hold
conversationownershipoperatingprofile_correction	correction and supersession lineage
conversationownershipoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Consent and Channel Capability — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-08-00
Subject	ConsentandChannelCapabilityChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Consent and Channel Capability
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Consent and Channel Capability.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Consent and Channel Capability.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConsentandChannelCapabilityChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConsentandChannelCapabilityChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConsentandChannelCapabilityChannel
- ValidateConsentandChannelCapabilityChannel
- ApproveConsentandChannelCapabilityChannel
- ActivateConsentandChannelCapabilityChannel
- SuspendConsentandChannelCapabilityChannel
- CorrectConsentandChannelCapabilityChannel
- RetireConsentandChannelCapabilityChannel

Reference Events:

- ConsentandChannelCapabilityChannelCreated
- ConsentandChannelCapabilityChannelValidated
- ConsentandChannelCapabilityChannelApproved
- ConsentandChannelCapabilityChannelActivated
- ConsentandChannelCapabilityChannelSuspended
- ConsentandChannelCapabilityChannelCorrected
- ConsentandChannelCapabilityChannelRetired

Reference queries:

- GetConsentandChannelCapabilityChannel
- ListConsentandChannelCapabilityChannelRecords
- GetConsentandChannelCapabilityChannelTimeline
- GetConsentandChannelCapabilityChannelEvidence
- GetConsentandChannelCapabilityChannelDecisionPackage
- GetConsentandChannelCapabilityChannelExceptions
- GetConsentandChannelCapabilityChannelAuditTrail

Reference API surface:

```
POST /api/v1/consentandchannelcapabilitychannel/commands
GET  /api/v1/consentandchannelcapabilitychannel/{id}
GET  /api/v1/consentandchannelcapabilitychannel/{id}/timeline
GET  /api/v1/consentandchannelcapabilitychannel/{id}/evidence
GET  /api/v1/consentandchannelcapabilitychannel/{id}/decisions
POST /api/v1/consentandchannelcapabilitychannel/{id}/exceptions
POST /api/v1/consentandchannelcapabilitychannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
consentandchannelcapabilitychannel_aggregate	authoritative subject
consentandchannelcapabilitychannel_version	immutable submitted, approved, issued, or signed versions
consentandchannelcapabilitychannel_relationship	governed relationships
consentandchannelcapabilitychannel_decision	binding Decisions and rationale
consentandchannelcapabilitychannel_evidence_link	provenance and evidence relationships
consentandchannelcapabilitychannel_event_outbox	transactional Event publication
consentandchannelcapabilitychannel_idempotency	duplicate-write protection
consentandchannelcapabilitychannel_exception	exception, claim, dispute, or hold
consentandchannelcapabilitychannel_correction	correction and supersession lineage
consentandchannelcapabilitychannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Consent and Channel Capability — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-08-01
Subject	ConsentandChannelCapabilityMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Consent and Channel Capability
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Consent and Channel Capability.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Consent and Channel Capability**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConsentandChannelCapabilityMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConsentandChannelCapabilityMessageModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConsentandChannelCapabilityMessageModel
- ValidateConsentandChannelCapabilityMessageModel
- ApproveConsentandChannelCapabilityMessageModel
- ActivateConsentandChannelCapabilityMessageModel
- SuspendConsentandChannelCapabilityMessageModel
- CorrectConsentandChannelCapabilityMessageModel
- RetireConsentandChannelCapabilityMessageModel

Reference Events:

- ConsentandChannelCapabilityMessageModelCreated
- ConsentandChannelCapabilityMessageModelValidated
- ConsentandChannelCapabilityMessageModelApproved
- ConsentandChannelCapabilityMessageModelActivated
- ConsentandChannelCapabilityMessageModelSuspended
- ConsentandChannelCapabilityMessageModelCorrected
- ConsentandChannelCapabilityMessageModelRetired

Reference queries:

- GetConsentandChannelCapabilityMessageModel
- ListConsentandChannelCapabilityMessageModelRecords
- GetConsentandChannelCapabilityMessageModelTimeline
- GetConsentandChannelCapabilityMessageModelEvidence
- GetConsentandChannelCapabilityMessageModelDecisionPackage
- GetConsentandChannelCapabilityMessageModelExceptions
- GetConsentandChannelCapabilityMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/consentandchannelcapabilitymessagemodel/commands
GET  /api/v1/consentandchannelcapabilitymessagemodel/{id}
GET  /api/v1/consentandchannelcapabilitymessagemodel/{id}/timeline
GET  /api/v1/consentandchannelcapabilitymessagemodel/{id}/evidence
GET  /api/v1/consentandchannelcapabilitymessagemodel/{id}/decisions
POST /api/v1/consentandchannelcapabilitymessagemodel/{id}/exceptions
POST /api/v1/consentandchannelcapabilitymessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
consentandchannelcapabilitymessagemodel_aggregate	authoritative subject
consentandchannelcapabilitymessagemodel_version	immutable submitted, approved, issued, or signed versions
consentandchannelcapabilitymessagemodel_relationship	governed relationships
consentandchannelcapabilitymessagemodel_decision	binding Decisions and rationale
consentandchannelcapabilitymessagemodel_evidence_link	provenance and evidence relationships
consentandchannelcapabilitymessagemodel_event_outbox	transactional Event publication
consentandchannelcapabilitymessagemodel_idempotency	duplicate-write protection
consentandchannelcapabilitymessagemodel_exception_correction	exception, claim, dispute, or hold correction and supersession lineage
consentandchannelcapabilitymessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.

Status	Meaning
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Consent and Channel Capability — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-08-02
Subject	ConsentandChannelCapabilityProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Consent and Channel Capability
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Consent and Channel Capability**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Consent and Channel Capability**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConsentandChannelCapabilityProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConsentandChannelCapabilityProviderAdapter {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConsentandChannelCapabilityProviderAdapter
- ValidateConsentandChannelCapabilityProviderAdapter
- ApproveConsentandChannelCapabilityProviderAdapter
- ActivateConsentandChannelCapabilityProviderAdapter
- SuspendConsentandChannelCapabilityProviderAdapter
- CorrectConsentandChannelCapabilityProviderAdapter
- RetireConsentandChannelCapabilityProviderAdapter

Reference Events:

- ConsentandChannelCapabilityProviderAdapterCreated
- ConsentandChannelCapabilityProviderAdapterValidated
- ConsentandChannelCapabilityProviderAdapterApproved
- ConsentandChannelCapabilityProviderAdapterActivated
- ConsentandChannelCapabilityProviderAdapterSuspended
- ConsentandChannelCapabilityProviderAdapterCorrected
- ConsentandChannelCapabilityProviderAdapterRetired

Reference queries:

- GetConsentandChannelCapabilityProviderAdapter
- ListConsentandChannelCapabilityProviderAdapterRecords
- GetConsentandChannelCapabilityProviderAdapterTimeline
- GetConsentandChannelCapabilityProviderAdapterEvidence
- GetConsentandChannelCapabilityProviderAdapterDecisionPackage
- GetConsentandChannelCapabilityProviderAdapterExceptions
- GetConsentandChannelCapabilityProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/consentandchannelcapabilityprovideradapter/commands
GET  /api/v1/consentandchannelcapabilityprovideradapter/{id}
GET  /api/v1/consentandchannelcapabilityprovideradapter/{id}/timeline
GET  /api/v1/consentandchannelcapabilityprovideradapter/{id}/evidence
GET  /api/v1/consentandchannelcapabilityprovideradapter/{id}/decisions
POST /api/v1/consentandchannelcapabilityprovideradapter/{id}/exceptions
POST /api/v1/consentandchannelcapabilityprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
consentandchannelcapabilityprovideradapter_ aggregate	authoritative subject
consentandchannelcapabilityprovideradapter_ version	immutable submitted, approved, issued, or signed versions
consentandchannelcapabilityprovideradapter_ relationship	governed relationships
consentandchannelcapabilityprovideradapter_ decision	binding Decisions and rationale
consentandchannelcapabilityprovideradapter_ evidence_link	provenance and evidence relationships
consentandchannelcapabilityprovideradapter_event_ outbox	transactional Event publication
consentandchannelcapabilityprovideradapter_ idempotency	duplicate-write protection
consentandchannelcapabilityprovideradapter_ exception	exception, claim, dispute, or hold
consentandchannelcapabilityprovideradapter_ correction	correction and supersession lineage
consentandchannelcapabilityprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Consent and Channel Capability — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-08-03
Subject	ConsentandChannelCapability0peratingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Consent and Channel Capability
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Consent and Channel Capability.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Consent and Channel Capability.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConsentandChannelCapabilityOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConsentandChannelCapabilityOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConsentandChannelCapabilityOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConsentandChannelCapabilityOperatingProfile
- ValidateConsentandChannelCapabilityOperatingProfile
- ApproveConsentandChannelCapabilityOperatingProfile
- ActivateConsentandChannelCapabilityOperatingProfile
- SuspendConsentandChannelCapabilityOperatingProfile
- CorrectConsentandChannelCapabilityOperatingProfile
- RetireConsentandChannelCapabilityOperatingProfile

Reference Events:

- ConsentandChannelCapabilityOperatingProfileCreated
- ConsentandChannelCapabilityOperatingProfileValidated
- ConsentandChannelCapabilityOperatingProfileApproved
- ConsentandChannelCapabilityOperatingProfileActivated
- ConsentandChannelCapabilityOperatingProfileSuspended
- ConsentandChannelCapabilityOperatingProfileCorrected
- ConsentandChannelCapabilityOperatingProfileRetired

Reference queries:

- GetConsentandChannelCapabilityOperatingProfile
- ListConsentandChannelCapabilityOperatingProfileRecords
- GetConsentandChannelCapabilityOperatingProfileTimeline
- GetConsentandChannelCapabilityOperatingProfileEvidence
- GetConsentandChannelCapabilityOperatingProfileDecisionPackage
- GetConsentandChannelCapabilityOperatingProfileExceptions
- GetConsentandChannelCapabilityOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/consentandchannelcapabilityoperatingprofile/commands
GET  /api/v1/consentandchannelcapabilityoperatingprofile/{id}
GET  /api/v1/consentandchannelcapabilityoperatingprofile/{id}/timeline
GET  /api/v1/consentandchannelcapabilityoperatingprofile/{id}/evidence
GET  /api/v1/consentandchannelcapabilityoperatingprofile/{id}/decisions
POST /api/v1/consentandchannelcapabilityoperatingprofile/{id}/exceptions
POST /api/v1/consentandchannelcapabilityoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
consentandchannelcapabilityoperatingprofile_ aggregate	authoritative subject
consentandchannelcapabilityoperatingprofile_ version	immutable submitted, approved, issued, or signed versions
consentandchannelcapabilityoperatingprofile_ relationship	governed relationships
consentandchannelcapabilityoperatingprofile_ decision	binding Decisions and rationale
consentandchannelcapabilityoperatingprofile_ evidence_link	provenance and evidence relationships
consentandchannelcapabilityoperatingprofile_ event_outbox	transactional Event publication
consentandchannelcapabilityoperatingprofile_ idempotency	duplicate-write protection
consentandchannelcapabilityoperatingprofile_ exception	exception, claim, dispute, or hold
consentandchannelcapabilityoperatingprofile_ correction	correction and supersession lineage
consentandchannelcapabilityoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Template and Localization — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-09-00
Subject	TemplateandLocalizationChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Template and Localization
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Template and Localization.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Template and Localization**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TemplateandLocalizationChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TemplateandLocalizationChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTemplateandLocalizationChannel
- ValidateTemplateandLocalizationChannel
- ApproveTemplateandLocalizationChannel
- ActivateTemplateandLocalizationChannel
- SuspendTemplateandLocalizationChannel
- CorrectTemplateandLocalizationChannel
- RetireTemplateandLocalizationChannel

Reference Events:

- TemplateandLocalizationChannelCreated
- TemplateandLocalizationChannelValidated
- TemplateandLocalizationChannelApproved
- TemplateandLocalizationChannelActivated
- TemplateandLocalizationChannelSuspended
- TemplateandLocalizationChannelCorrected
- TemplateandLocalizationChannelRetired

Reference queries:

- GetTemplateandLocalizationChannel
- ListTemplateandLocalizationChannelRecords
- GetTemplateandLocalizationChannelTimeline
- GetTemplateandLocalizationChannelEvidence
- GetTemplateandLocalizationChannelDecisionPackage
- GetTemplateandLocalizationChannelExceptions
- GetTemplateandLocalizationChannelAuditTrail

Reference API surface:

```
POST /api/v1/templateandlocalizationchannel/commands
GET  /api/v1/templateandlocalizationchannel/{id}
GET  /api/v1/templateandlocalizationchannel/{id}/timeline
GET  /api/v1/templateandlocalizationchannel/{id}/evidence
GET  /api/v1/templateandlocalizationchannel/{id}/decisions
POST /api/v1/templateandlocalizationchannel/{id}/exceptions
POST /api/v1/templateandlocalizationchannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
templateandlocalizationchannel_aggregate	authoritative subject
templateandlocalizationchannel_version	immutable submitted, approved, issued, or signed versions
templateandlocalizationchannel_relationship	governed relationships
templateandlocalizationchannel_decision	binding Decisions and rationale
templateandlocalizationchannel_evidence_link	provenance and evidence relationships
templateandlocalizationchannel_event_outbox	transactional Event publication
templateandlocalizationchannel_idempotency	duplicate-write protection
templateandlocalizationchannel_exception	exception, claim, dispute, or hold
templateandlocalizationchannel_correction	correction and supersession lineage
templateandlocalizationchannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Template and Localization — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-09-01
Subject	TemplateandLocalizationMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Template and Localization
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Template and Localization.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Template and Localization**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TemplateandLocalizationMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TemplateandLocalizationMessageModel {  
  subjectId  
  tenantId  
  organizationContext  
  sourceIdentityRefs[]  
  sourceVersionRefs[]  
  lifecycleState  
  relationshipRefs[]  
  decisionRefs[]  
  intentRefs[]  
  evidenceSetRef  
  knowledgeCutoff  
  policyRefs[]  
  authorityRef  
  jurisdictionRefs[]  
  createdAt  
  effectiveAt  
  updatedAt  
  version  
  correctionOf?  
  correlationId  
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTemplateandLocalizationMessageModel
- ValidateTemplateandLocalizationMessageModel
- ApproveTemplateandLocalizationMessageModel
- ActivateTemplateandLocalizationMessageModel
- SuspendTemplateandLocalizationMessageModel
- CorrectTemplateandLocalizationMessageModel
- RetireTemplateandLocalizationMessageModel

Reference Events:

- TemplateandLocalizationMessageModelCreated
- TemplateandLocalizationMessageModelValidated
- TemplateandLocalizationMessageModelApproved
- TemplateandLocalizationMessageModelActivated
- TemplateandLocalizationMessageModelSuspended
- TemplateandLocalizationMessageModelCorrected
- TemplateandLocalizationMessageModelRetired

Reference queries:

- GetTemplateandLocalizationMessageModel
- ListTemplateandLocalizationMessageModelRecords
- GetTemplateandLocalizationMessageModelTimeline
- GetTemplateandLocalizationMessageModelEvidence
- GetTemplateandLocalizationMessageModelDecisionPackage
- GetTemplateandLocalizationMessageModelExceptions
- GetTemplateandLocalizationMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/templateandlocalizationmessagemodel/commands
GET  /api/v1/templateandlocalizationmessagemodel/{id}
GET  /api/v1/templateandlocalizationmessagemodel/{id}/timeline
GET  /api/v1/templateandlocalizationmessagemodel/{id}/evidence
GET  /api/v1/templateandlocalizationmessagemodel/{id}/decisions
POST /api/v1/templateandlocalizationmessagemodel/{id}/exceptions
POST /api/v1/templateandlocalizationmessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
templateandlocalizationmessagemodel_aggregate	authoritative subject
templateandlocalizationmessagemodel_version	immutable submitted, approved, issued, or signed versions
templateandlocalizationmessagemodel_relationship	governed relationships
templateandlocalizationmessagemodel_decision	binding Decisions and rationale
templateandlocalizationmessagemodel_evidence_link	provenance and evidence relationships
templateandlocalizationmessagemodel_event_outbox	transactional Event publication
templateandlocalizationmessagemodel_idempotency	duplicate-write protection
templateandlocalizationmessagemodel_exception	exception, claim, dispute, or hold
templateandlocalizationmessagemodel_correction	correction and supersession lineage
templateandlocalizationmessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Template and Localization — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-09-02
Subject	TemplateandLocalizationProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Template and Localization
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Template and Localization**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Template and Localization**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TemplateandLocalizationProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TemplateandLocalizationProviderAdapter {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTemplateandLocalizationProviderAdapter
- ValidateTemplateandLocalizationProviderAdapter
- ApproveTemplateandLocalizationProviderAdapter
- ActivateTemplateandLocalizationProviderAdapter
- SuspendTemplateandLocalizationProviderAdapter
- CorrectTemplateandLocalizationProviderAdapter
- RetireTemplateandLocalizationProviderAdapter

Reference Events:

- TemplateandLocalizationProviderAdapterCreated
- TemplateandLocalizationProviderAdapterValidated
- TemplateandLocalizationProviderAdapterApproved
- TemplateandLocalizationProviderAdapterActivated
- TemplateandLocalizationProviderAdapterSuspended
- TemplateandLocalizationProviderAdapterCorrected
- TemplateandLocalizationProviderAdapterRetired

Reference queries:

- GetTemplateandLocalizationProviderAdapter
- ListTemplateandLocalizationProviderAdapterRecords
- GetTemplateandLocalizationProviderAdapterTimeline
- GetTemplateandLocalizationProviderAdapterEvidence
- GetTemplateandLocalizationProviderAdapterDecisionPackage
- GetTemplateandLocalizationProviderAdapterExceptions
- GetTemplateandLocalizationProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/templateandlocalizationprovideradapter/commands
GET  /api/v1/templateandlocalizationprovideradapter/{id}
GET  /api/v1/templateandlocalizationprovideradapter/{id}/timeline
GET  /api/v1/templateandlocalizationprovideradapter/{id}/evidence
GET  /api/v1/templateandlocalizationprovideradapter/{id}/decisions
POST /api/v1/templateandlocalizationprovideradapter/{id}/exceptions
POST /api/v1/templateandlocalizationprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
templateandlocalizationprovideradapter_aggregate	authoritative subject
templateandlocalizationprovideradapter_version	immutable submitted, approved, issued, or signed versions
templateandlocalizationprovideradapter_relationship	governed relationships
templateandlocalizationprovideradapter_decision	binding Decisions and rationale
templateandlocalizationprovideradapter_evidence_link	provenance and evidence relationships
templateandlocalizationprovideradapter_event_outbox	transactional Event publication
templateandlocalizationprovideradapter_idempotency	duplicate-write protection
templateandlocalizationprovideradapter_exception	exception, claim, dispute, or hold
templateandlocalizationprovideradapter_correction	correction and supersession lineage
templateandlocalizationprovideradapter_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Template and Localization — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-09-03
Subject	TemplateandLocalizationOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Template and Localization
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Template and Localization.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Template and Localization**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TemplateandLocalizationOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TemplateandLocalizationOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TemplateandLocalizationOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTemplateandLocalizationOperatingProfile
- ValidateTemplateandLocalizationOperatingProfile
- ApproveTemplateandLocalizationOperatingProfile
- ActivateTemplateandLocalizationOperatingProfile
- SuspendTemplateandLocalizationOperatingProfile
- CorrectTemplateandLocalizationOperatingProfile
- RetireTemplateandLocalizationOperatingProfile

Reference Events:

- TemplateandLocalizationOperatingProfileCreated
- TemplateandLocalizationOperatingProfileValidated
- TemplateandLocalizationOperatingProfileApproved
- TemplateandLocalizationOperatingProfileActivated
- TemplateandLocalizationOperatingProfileSuspended
- TemplateandLocalizationOperatingProfileCorrected
- TemplateandLocalizationOperatingProfileRetired

Reference queries:

- GetTemplateandLocalizationOperatingProfile
- ListTemplateandLocalizationOperatingProfileRecords
- GetTemplateandLocalizationOperatingProfileTimeline
- GetTemplateandLocalizationOperatingProfileEvidence
- GetTemplateandLocalizationOperatingProfileDecisionPackage
- GetTemplateandLocalizationOperatingProfileExceptions
- GetTemplateandLocalizationOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/templateandlocalizationoperatingprofile/commands
GET  /api/v1/templateandlocalizationoperatingprofile/{id}
GET  /api/v1/templateandlocalizationoperatingprofile/{id}/timeline
GET  /api/v1/templateandlocalizationoperatingprofile/{id}/evidence
GET  /api/v1/templateandlocalizationoperatingprofile/{id}/decisions
POST /api/v1/templateandlocalizationoperatingprofile/{id}/exceptions
POST /api/v1/templateandlocalizationoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
templateandlocalizationoperatingprofile_aggregate	authoritative subject
templateandlocalizationoperatingprofile_version	immutable submitted, approved, issued, or signed versions
templateandlocalizationoperatingprofile_relationship	governed relationships
templateandlocalizationoperatingprofile_decision	binding Decisions and rationale
templateandlocalizationoperatingprofile_evidence_link	provenance and evidence relationships
templateandlocalizationoperatingprofile_event_outbox	transactional Event publication
templateandlocalizationoperatingprofile_idempotency	duplicate-write protection
templateandlocalizationoperatingprofile_exception	exception, claim, dispute, or hold
templateandlocalizationoperatingprofile_correction	correction and supersession lineage
templateandlocalizationoperatingprofile_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.

Status	Meaning
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Human Handoff and AI-Assisted Response — Orientation, Ownership, Consent, and Channel Capability

Metadata	Value
Volume	8
Chapter ID	V08-10-00
Subject	HumanHandoffandAI-AssistedResponseChannel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Human Handoff and AI-Assisted Response
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define ownership, participant identity, consent, channel capability, provider boundary, retention, and audit for Human Handoff and AI-Assisted Response.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Human Handoff and AI-Assisted Response.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
HumanHandoffandAI-AssistedResponseChannel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseChannelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseChannelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseChannelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseChannelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
HumanHandoffandAI-AssistedResponseChannel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

- A conformant design distinguishes:
- the natural person acting;
 - the service or AI identity invoking a Tool;
 - the organization or principal represented;
 - the role or relationship granting context;
 - the mandate or delegation granting action authority;
 - the Policy and jurisdiction limiting the action;
 - the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateHumanHandoffandAI-AssistedResponseChannel
- ValidateHumanHandoffandAI-AssistedResponseChannel
- ApproveHumanHandoffandAI-AssistedResponseChannel
- ActivateHumanHandoffandAI-AssistedResponseChannel
- SuspendHumanHandoffandAI-AssistedResponseChannel
- CorrectHumanHandoffandAI-AssistedResponseChannel
- RetireHumanHandoffandAI-AssistedResponseChannel

Reference Events:

- HumanHandoffandAI-AssistedResponseChannelCreated
- HumanHandoffandAI-AssistedResponseChannelValidated
- HumanHandoffandAI-AssistedResponseChannelApproved
- HumanHandoffandAI-AssistedResponseChannelActivated
- HumanHandoffandAI-AssistedResponseChannelSuspended
- HumanHandoffandAI-AssistedResponseChannelCorrected
- HumanHandoffandAI-AssistedResponseChannelRetired

Reference queries:

- GetHumanHandoffandAI-AssistedResponseChannel
- ListHumanHandoffandAI-AssistedResponseChannelRecords
- GetHumanHandoffandAI-AssistedResponseChannelTimeline
- GetHumanHandoffandAI-AssistedResponseChannelEvidence
- GetHumanHandoffandAI-AssistedResponseChannelDecisionPackage
- GetHumanHandoffandAI-AssistedResponseChannelExceptions
- GetHumanHandoffandAI-AssistedResponseChannelAuditTrail

Reference API surface:

```
POST /api/v1/humanhandoffandai_assistedresponsechannel/commands
GET  /api/v1/humanhandoffandai_assistedresponsechannel/{id}
GET  /api/v1/humanhandoffandai_assistedresponsechannel/{id}/timeline
GET  /api/v1/humanhandoffandai_assistedresponsechannel/{id}/evidence
GET  /api/v1/humanhandoffandai_assistedresponsechannel/{id}/decisions
POST /api/v1/humanhandoffandai_assistedresponsechannel/{id}/exceptions
POST /api/v1/humanhandoffandai_assistedresponsechannel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
humanhandoffandai_assistedresponsechannel_ aggregate	authoritative subject
humanhandoffandai_assistedresponsechannel_version humanhandoffandai_assistedresponsechannel_ relationship	immutable submitted, approved, issued, or signed versions governed relationships
humanhandoffandai_assistedresponsechannel_ decision	binding Decisions and rationale
humanhandoffandai_assistedresponsechannel_ evidence_link	provenance and evidence relationships
humanhandoffandai_assistedresponsechannel_event_ outbox	transactional Event publication
humanhandoffandai_assistedresponsechannel_ idempotency	duplicate-write protection
humanhandoffandai_assistedresponsechannel_ exception	exception, claim, dispute, or hold
humanhandoffandai_assistedresponsechannel_ correction	correction and supersession lineage
humanhandoffandai_assistedresponsechannel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.

- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
IMPLEMENTED_WITH_GAP	
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.

Status	Meaning
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Human Handoff and AI-Assisted Response — Messages, Conversations, Templates, and State

Metadata	Value
Volume	8
Chapter ID	V08-10-01
Subject	HumanHandoffandAI-AssistedResponseMessageModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Human Handoff and AI-Assisted Response
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define message, conversation, template, localization, delivery, read, failure, correction, and handoff state for Human Handoff and AI-Assisted Response.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Human Handoff and AI-Assisted Response.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
HumanHandoffandAI-AssistedResponseMessageModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseMessageModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseMessageModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseMessageModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseMessageModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
HumanHandoffandAI-AssistedResponseMessageModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions

- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateHumanHandoffandAI-AssistedResponseMessageModel
- ValidateHumanHandoffandAI-AssistedResponseMessageModel
- ApproveHumanHandoffandAI-AssistedResponseMessageModel
- ActivateHumanHandoffandAI-AssistedResponseMessageModel
- SuspendHumanHandoffandAI-AssistedResponseMessageModel
- CorrectHumanHandoffandAI-AssistedResponseMessageModel
- RetireHumanHandoffandAI-AssistedResponseMessageModel

Reference Events:

- HumanHandoffandAI-AssistedResponseMessageModelCreated
- HumanHandoffandAI-AssistedResponseMessageModelValidated
- HumanHandoffandAI-AssistedResponseMessageModelApproved
- HumanHandoffandAI-AssistedResponseMessageModelActivated
- HumanHandoffandAI-AssistedResponseMessageModelSuspended
- HumanHandoffandAI-AssistedResponseMessageModelCorrected
- HumanHandoffandAI-AssistedResponseMessageModelRetired

Reference queries:

- GetHumanHandoffandAI-AssistedResponseMessageModel
- ListHumanHandoffandAI-AssistedResponseMessageModelRecords
- GetHumanHandoffandAI-AssistedResponseMessageModelTimeline
- GetHumanHandoffandAI-AssistedResponseMessageModelEvidence
- GetHumanHandoffandAI-AssistedResponseMessageModelDecisionPackage
- GetHumanHandoffandAI-AssistedResponseMessageModelExceptions
- GetHumanHandoffandAI-AssistedResponseMessageModelAuditTrail

Reference API surface:

```
POST /api/v1/humanhandoffandai_assistedresponsemessagemodel/commands
GET  /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}
GET  /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}/timeline
GET  /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}/evidence
GET  /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}/decisions
POST /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}/exceptions
POST /api/v1/humanhandoffandai_assistedresponsemessagemodel/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
humanhandoffandai_assistedresponsemessagemodel_aggregate	authoritative subject
humanhandoffandai_assistedresponsemessagemodel_version	immutable submitted, approved, issued, or signed versions
humanhandoffandai_assistedresponsemessagemodel_relationship	governed relationships
humanhandoffandai_assistedresponsemessagemodel_decision	binding Decisions and rationale
humanhandoffandai_assistedresponsemessagemodel_evidence_link	provenance and evidence relationships
humanhandoffandai_assistedresponsemessagemodel_event_outbox	transactional Event publication
humanhandoffandai_assistedresponsemessagemodel_idempotency	duplicate-write protection
humanhandoffandai_assistedresponsemessagemodel_exception	exception, claim, dispute, or hold
humanhandoffandai_assistedresponsemessagemodel_correction	correction and supersession lineage
humanhandoffandai_assistedresponsemessagemodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Human Handoff and AI-Assisted Response — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation

Metadata	Value
Volume	8
Chapter ID	V08-10-02
Subject	HumanHandoffandAI-AssistedResponseProviderAdapter
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Human Handoff and AI-Assisted Response
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define provider request, callback, signature, idempotency, ordering, retry, outage, Outcome Unknown, DLQ and reconciliation for Human Handoff and AI-Assisted Response**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Human Handoff and AI-Assisted Response**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events

- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
HumanHandoffandAI-AssistedResponseProviderAdapter	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseProviderAdapterVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseProviderAdapterDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseProviderAdapterEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseProviderAdapterException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
HumanHandoffandAI-AssistedResponseProviderAdapter {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
 - typed contracts
 - immutable binding versions
 - transactional outbox
 - idempotent Commands
 - explicit Outcome Unknown
 - correction lineage
 - machine-verifiable conformance
-

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateHumanHandoffandAI-AssistedResponseProviderAdapter
- ValidateHumanHandoffandAI-AssistedResponseProviderAdapter
- ApproveHumanHandoffandAI-AssistedResponseProviderAdapter
- ActivateHumanHandoffandAI-AssistedResponseProviderAdapter
- SuspendHumanHandoffandAI-AssistedResponseProviderAdapter
- CorrectHumanHandoffandAI-AssistedResponseProviderAdapter
- RetireHumanHandoffandAI-AssistedResponseProviderAdapter

Reference Events:

- HumanHandoffandAI-AssistedResponseProviderAdapterCreated
- HumanHandoffandAI-AssistedResponseProviderAdapterValidated
- HumanHandoffandAI-AssistedResponseProviderAdapterApproved
- HumanHandoffandAI-AssistedResponseProviderAdapterActivated
- HumanHandoffandAI-AssistedResponseProviderAdapterSuspended
- HumanHandoffandAI-AssistedResponseProviderAdapterCorrected
- HumanHandoffandAI-AssistedResponseProviderAdapterRetired

Reference queries:

- GetHumanHandoffandAI-AssistedResponseProviderAdapter
- ListHumanHandoffandAI-AssistedResponseProviderAdapterRecords
- GetHumanHandoffandAI-AssistedResponseProviderAdapterTimeline
- GetHumanHandoffandAI-AssistedResponseProviderAdapterEvidence
- GetHumanHandoffandAI-AssistedResponseProviderAdapterDecisionPackage
- GetHumanHandoffandAI-AssistedResponseProviderAdapterExceptions
- GetHumanHandoffandAI-AssistedResponseProviderAdapterAuditTrail

Reference API surface:

```
POST /api/v1/humanhandoffandai_assistedresponseprovideradapter/commands
GET  /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}
GET  /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}/timeline
GET  /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}/evidence
GET  /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}/decisions
POST /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}/exceptions
POST /api/v1/humanhandoffandai_assistedresponseprovideradapter/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
humanhandoffandai_	authoritative subject
assistedresponseprovideradapter_aggregate	
humanhandoffandai_	immutable submitted, approved, issued, or signed versions
assistedresponseprovideradapter_version	
humanhandoffandai_	governed relationships
assistedresponseprovideradapter_relationship	
humanhandoffandai_	binding Decisions and rationale
assistedresponseprovideradapter_decision	
humanhandoffandai_	provenance and evidence relationships
assistedresponseprovideradapter_evidence_link	
humanhandoffandai_	transactional Event publication
assistedresponseprovideradapter_event_outbox	
humanhandoffandai_	duplicate-write protection
assistedresponseprovideradapter_idempotency	
humanhandoffandai_	exception, claim, dispute, or hold
assistedresponseprovideradapter_exception	
humanhandoffandai_	correction and supersession lineage
assistedresponseprovideradapter_correction	
humanhandoffandai_	privileged action evidence
assistedresponseprovideradapter_audit	

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Human Handoff and AI-Assisted Response — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests

Metadata	Value
Volume	8
Chapter ID	V08-10-03
Subject	HumanHandoffandAI-AssistedResponseOperatingProfile
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Architecture / Human Handoff and AI-Assisted Response
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define Portal experience, CRM linkage, AI drafting, human handoff, security, SLOs, operations, tests and conformance for Human Handoff and AI-Assisted Response.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Architecture / Human Handoff and AI-Assisted Response.** Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands

- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
HumanHandoffandAI-AssistedResponseOperatingProfile	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseOperatingProfileVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseOperatingProfileDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseOperatingProfileEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
HumanHandoffandAI-AssistedResponseOperatingProfileException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
HumanHandoffandAI-AssistedResponseOperatingProfile {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
 - typed contracts
 - immutable binding versions
 - transactional outbox
 - idempotent Commands
 - explicit Outcome Unknown
 - correction lineage
 - machine-verifiable conformance
-

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateHumanHandoffandAI-AssistedResponseOperatingProfile
- ValidateHumanHandoffandAI-AssistedResponseOperatingProfile
- ApproveHumanHandoffandAI-AssistedResponseOperatingProfile
- ActivateHumanHandoffandAI-AssistedResponseOperatingProfile
- SuspendHumanHandoffandAI-AssistedResponseOperatingProfile
- CorrectHumanHandoffandAI-AssistedResponseOperatingProfile
- RetireHumanHandoffandAI-AssistedResponseOperatingProfile

Reference Events:

- HumanHandoffandAI-AssistedResponseOperatingProfileCreated
- HumanHandoffandAI-AssistedResponseOperatingProfileValidated
- HumanHandoffandAI-AssistedResponseOperatingProfileApproved
- HumanHandoffandAI-AssistedResponseOperatingProfileActivated
- HumanHandoffandAI-AssistedResponseOperatingProfileSuspended
- HumanHandoffandAI-AssistedResponseOperatingProfileCorrected
- HumanHandoffandAI-AssistedResponseOperatingProfileRetired

Reference queries:

- GetHumanHandoffandAI-AssistedResponseOperatingProfile
- ListHumanHandoffandAI-AssistedResponseOperatingProfileRecords
- GetHumanHandoffandAI-AssistedResponseOperatingProfileTimeline
- GetHumanHandoffandAI-AssistedResponseOperatingProfileEvidence
- GetHumanHandoffandAI-AssistedResponseOperatingProfileDecisionPackage
- GetHumanHandoffandAI-AssistedResponseOperatingProfileExceptions
- GetHumanHandoffandAI-AssistedResponseOperatingProfileAuditTrail

Reference API surface:

```
POST /api/v1/humanhandoffandai_assistedresponseoperatingprofile/commands
GET  /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}
GET  /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}/timeline
GET  /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}/evidence
GET  /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}/decisions
POST /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}/exceptions
POST /api/v1/humanhandoffandai_assistedresponseoperatingprofile/{id}/corrections
```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]
```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
humanhandoffandai_	authoritative subject
assistedresponseoperatingprofile_aggregate	immutable submitted, approved, issued, or signed versions
humanhandoffandai_	governed relationships
assistedresponseoperatingprofile_relationship	binding Decisions and rationale
humanhandoffandai_	provenance and evidence relationships
assistedresponseoperatingprofile_evidence_link	transactional Event publication
humanhandoffandai_	duplicate-write protection
assistedresponseoperatingprofile_idempotency	exception, claim, dispute, or hold
humanhandoffandai_	correction and supersession lineage
assistedresponseoperatingprofile_correction	privileged action evidence
humanhandoffandai_	
assistedresponseoperatingprofile_audit	

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.

- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- draft responses with approved templates and sources
- classify intent and urgency
- summarize conversation history
- recommend human handoff without impersonating the human

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests

- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.

Status	Meaning
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Engagement Channel Ownership and Consent Matrix

Metadata	Value
Volume	8
Chapter ID	V08-SYNTH
Subject	ChannelOwnershipMatrix
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Engagement Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **connect channels, providers, consent, templates, conversations, delivery evidence, CRM, AI assistance, human handoff, retention, and audit.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Engagement Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ChannelOwnershipMatrix	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ChannelOwnershipMatrixVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ChannelOwnershipMatrixDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ChannelOwnershipMatrixEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ChannelOwnershipMatrixException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ChannelOwnershipMatrix {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```



```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateChannelOwnershipMatrix
- ValidateChannelOwnershipMatrix
- ApproveChannelOwnershipMatrix
- ActivateChannelOwnershipMatrix
- SuspendChannelOwnershipMatrix
- CorrectChannelOwnershipMatrix
- RetireChannelOwnershipMatrix

Reference Events:

- ChannelOwnershipMatrixCreated
- ChannelOwnershipMatrixValidated
- ChannelOwnershipMatrixApproved
- ChannelOwnershipMatrixActivated
- ChannelOwnershipMatrixSuspended
- ChannelOwnershipMatrixCorrected
- ChannelOwnershipMatrixRetired

Reference queries:

- GetChannelOwnershipMatrix
- ListChannelOwnershipMatrixRecords
- GetChannelOwnershipMatrixTimeline
- GetChannelOwnershipMatrixEvidence
- GetChannelOwnershipMatrixDecisionPackage
- GetChannelOwnershipMatrixExceptions
- GetChannelOwnershipMatrixAuditTrail

Reference API surface:

```

POST /api/v1/channelownershipmatrix/commands
GET  /api/v1/channelownershipmatrix/{id}
GET  /api/v1/channelownershipmatrix/{id}/timeline
GET  /api/v1/channelownershipmatrix/{id}/evidence
GET  /api/v1/channelownershipmatrix/{id}/decisions
POST /api/v1/channelownershipmatrix/{id}/exceptions
POST /api/v1/channelownershipmatrix/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
channelownershipmatrix_aggregate	authoritative subject
channelownershipmatrix_version	immutable submitted, approved, issued, or signed versions
channelownershipmatrix_relationship	governed relationships
channelownershipmatrix_decision	binding Decisions and rationale
channelownershipmatrix_evidence_link	provenance and evidence relationships
channelownershipmatrix_event_outbox	transactional Event publication
channelownershipmatrix_idempotency	duplicate-write protection
channelownershipmatrix_exception	exception, claim, dispute, or hold
channelownershipmatrix_correction	correction and supersession lineage
channelownershipmatrix_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions

- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.